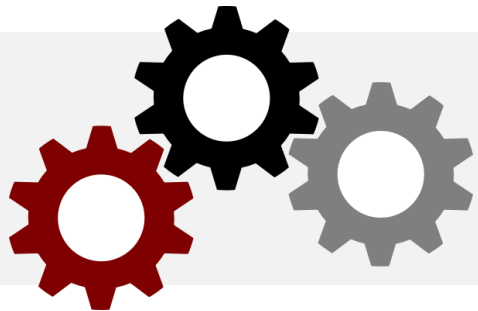

SQL Server

Always On Availability Groups

and

FAILOVER CLUSTERING



INTERNALS

EDWIN SARMIENTO

~~Microsoft MVP~~/Microsoft Certified Master: SQL Server



[https://**LearnSQLServerHADR**.com](https://LearnSQLServerHADR.com)



edwin@EdwinMSarmiento.com



[@EdwinMSarmiento](https://twitter.com/EdwinMSarmiento)



<https://ca.linkedin.com/in/EdwinMSarmiento>



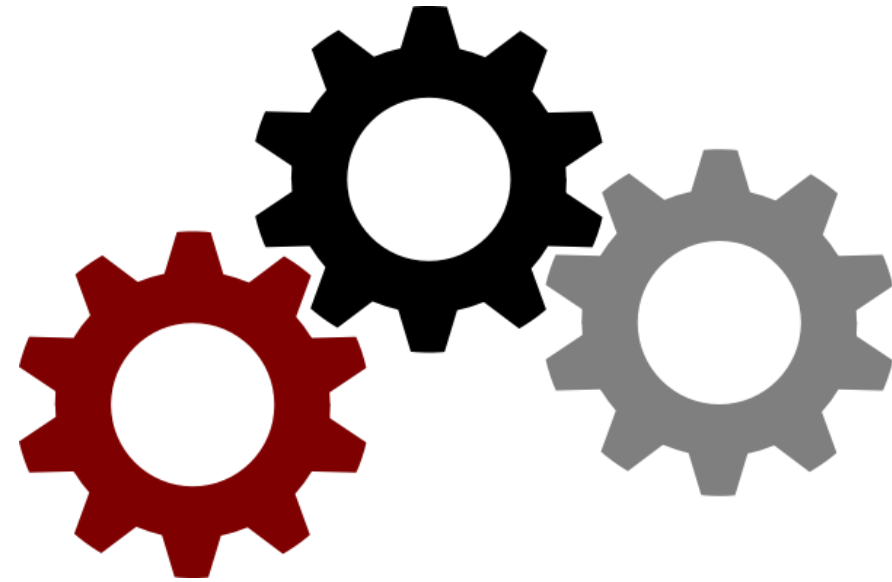
<https://www.youtube.com/@LearnSQLServerHADR>



QUICK
SURVEY

HOW HIGH AVAILABILITY WORKS IN

SQL Server



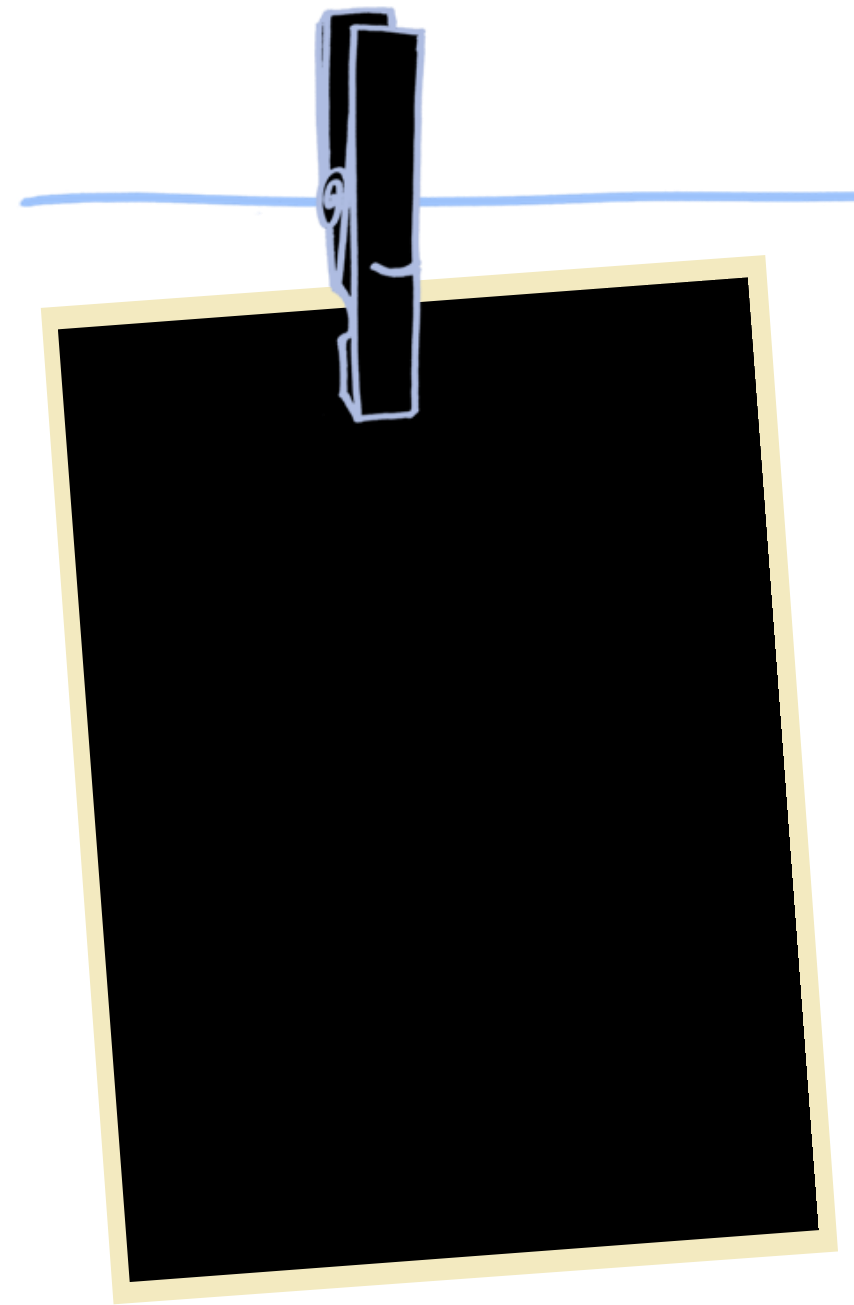




AIR TRAFFIC CONTROL



CONSEQUENCES OF **MISCOMMUNICATION**



SOURCE: CCTV video of the Air Canada accident at LaGuardia

SQL Server

RELIES ON ANOTHER PLATFORM FOR
HIGH AVAILABILITY

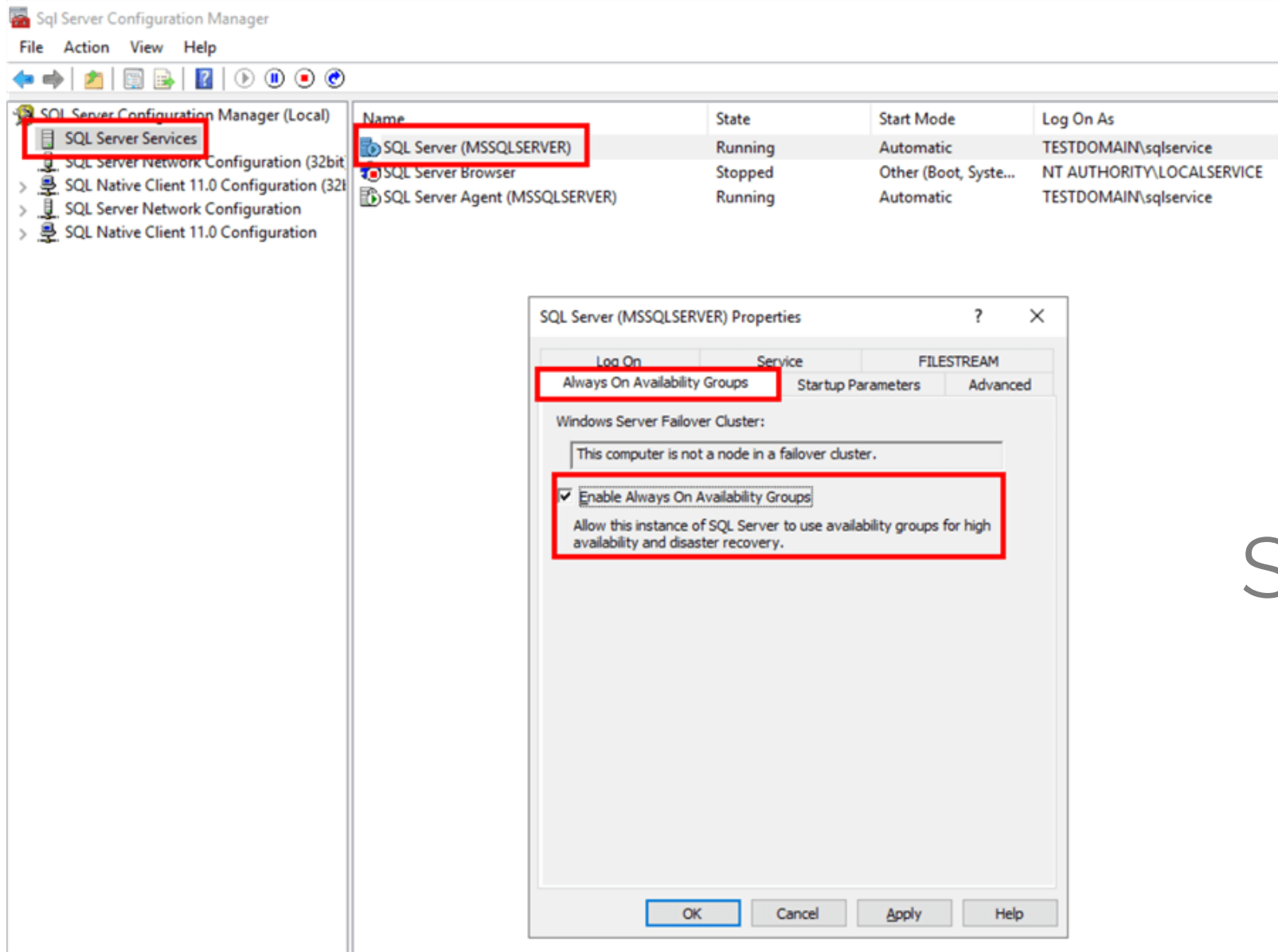
WINDOWS SERVER

FAILOVER CLUSTER

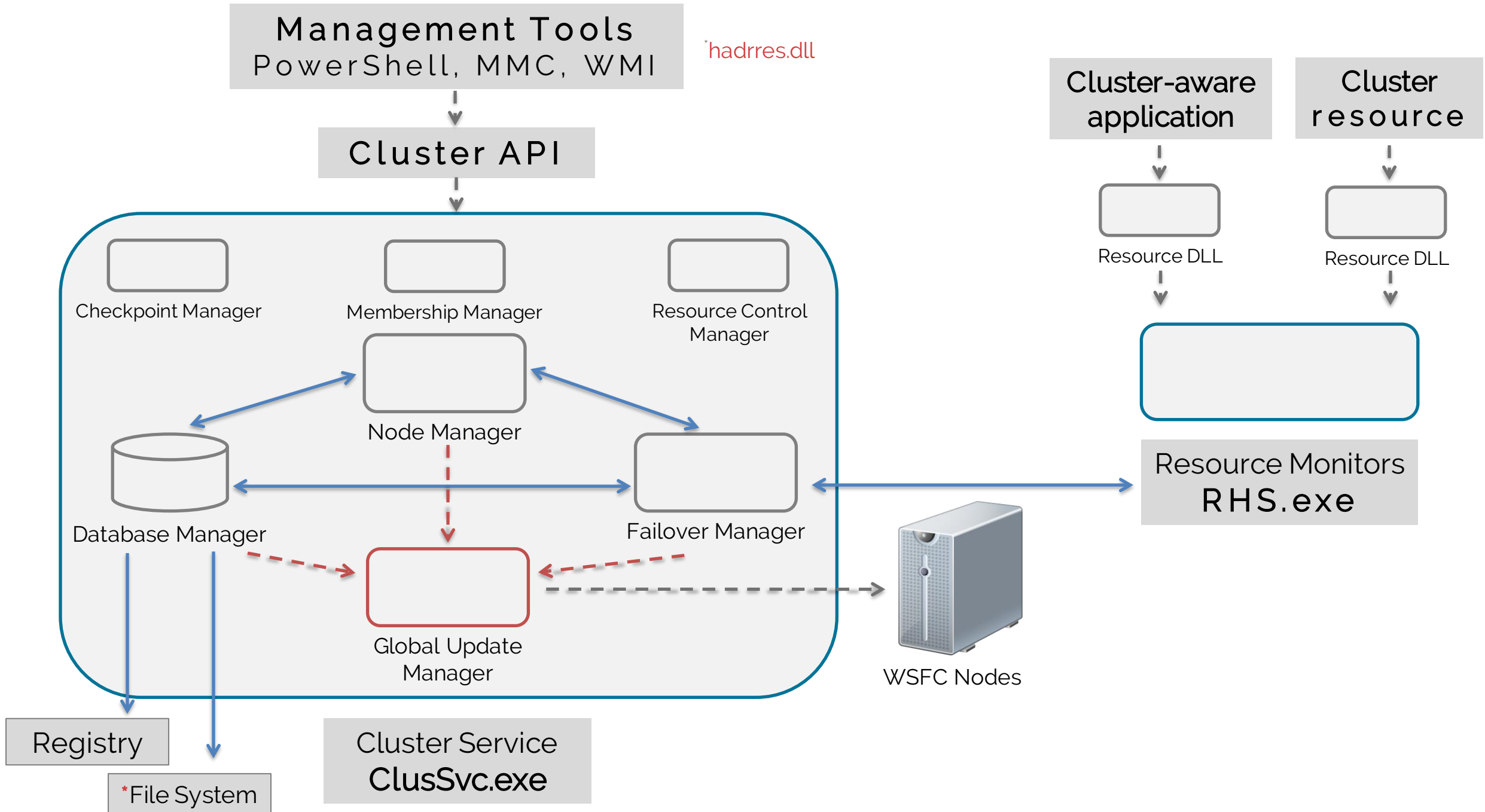
WSFC INTERNALS

because ^{highly available} **AVAILABILITY GROUPS**
run on top of a WSFC

* understanding **WSFC internals** in the context of
ADMINISTRATION & TROUBLESHOOTING



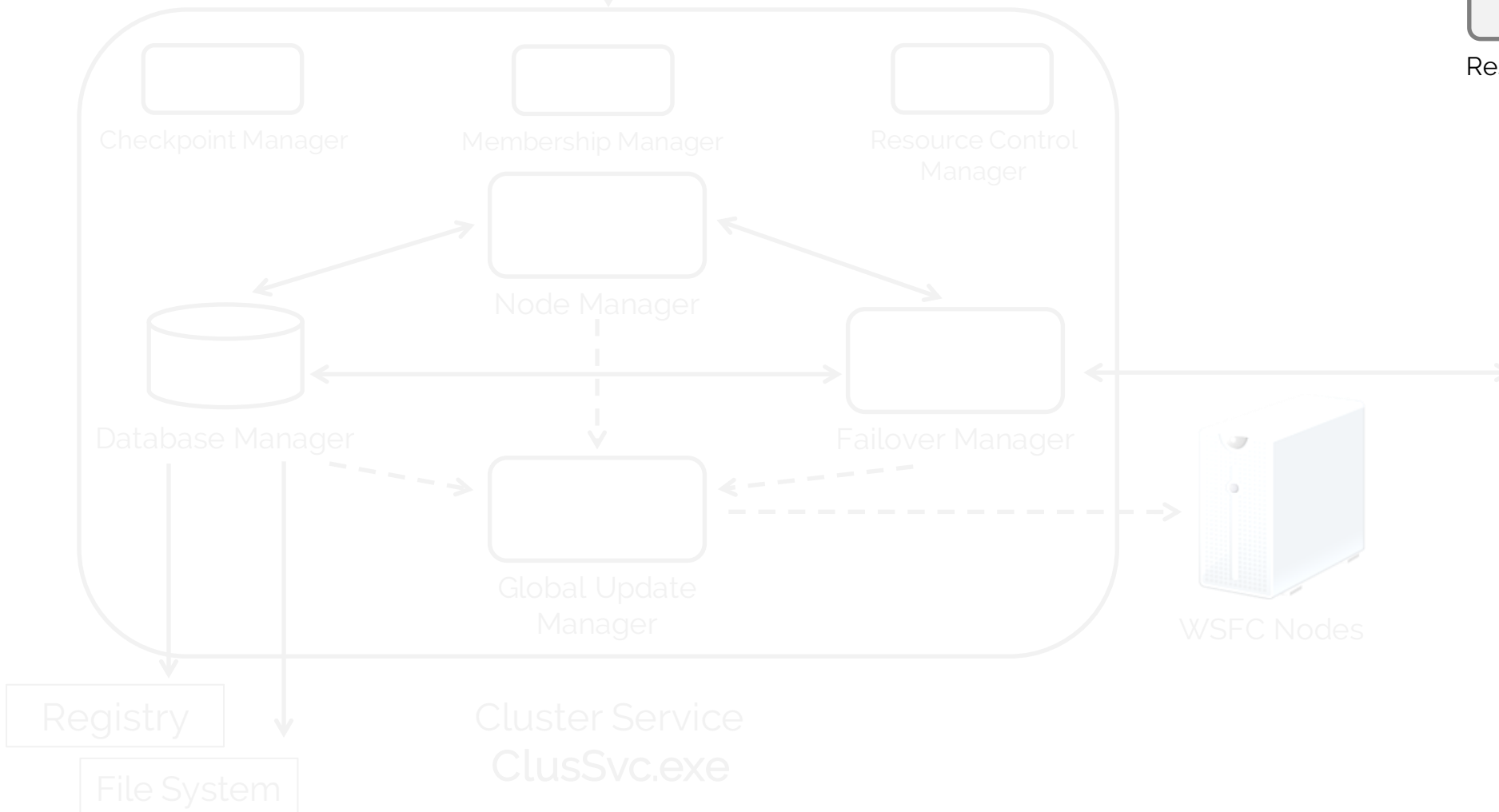
*starting with
SQL Server 2017



Management Tools

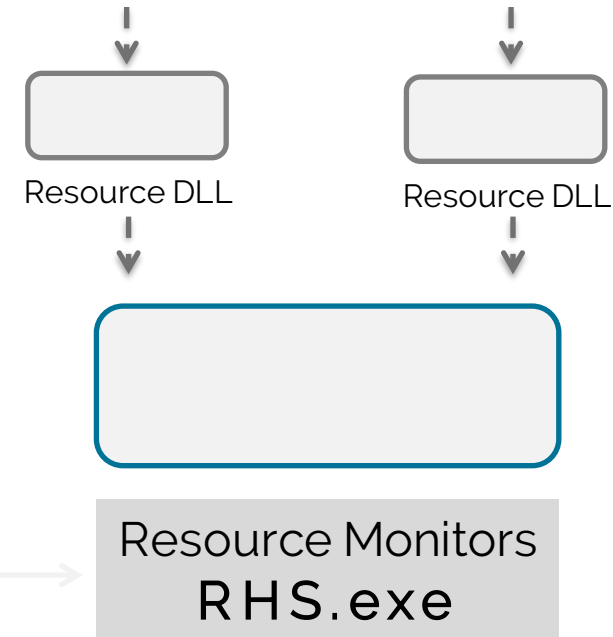
PowerShell, MMC, WMI

Cluster API



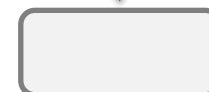
Cluster-aware application

Cluster resource

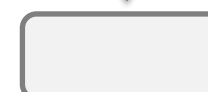


Cluster-aware
application

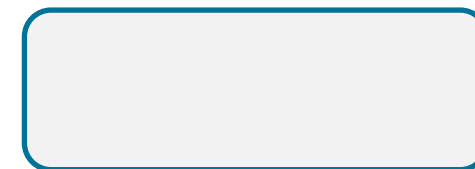
Cluster
resource



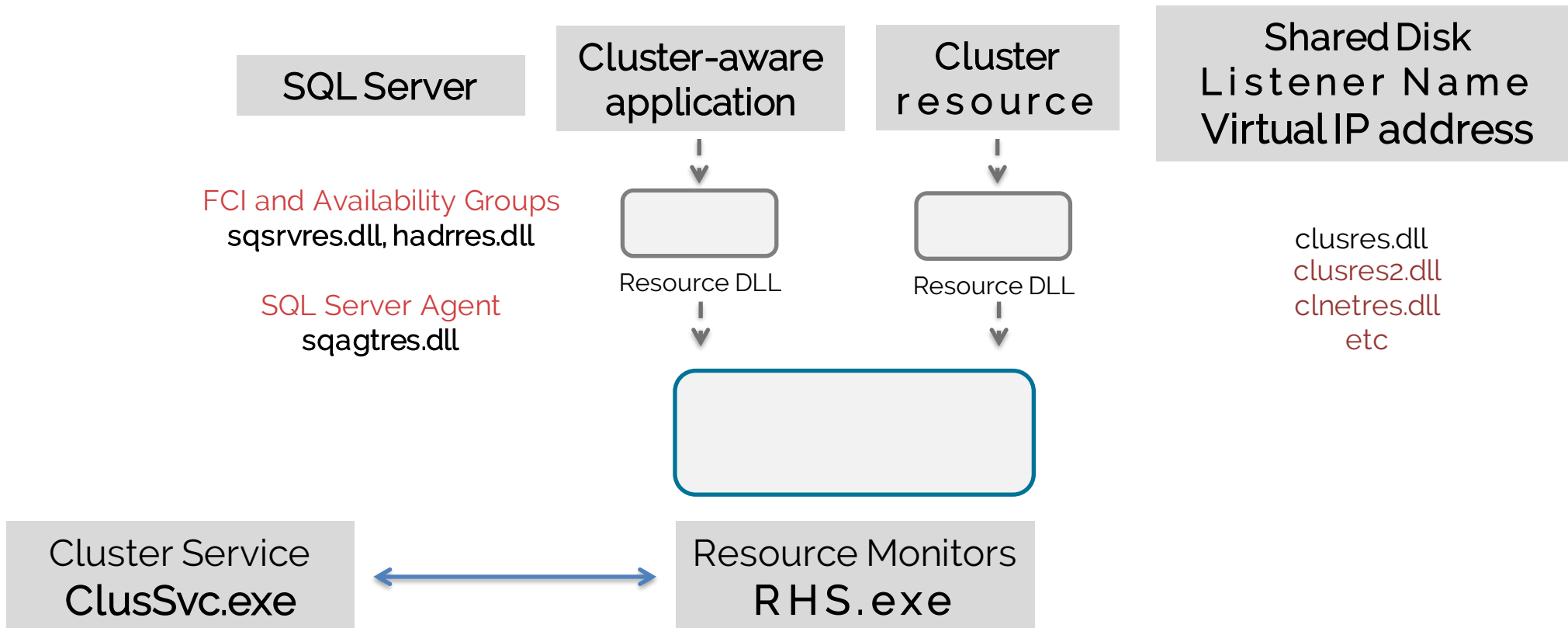
Resource DLL



Resource DLL



Resource Monitors
RHS.exe

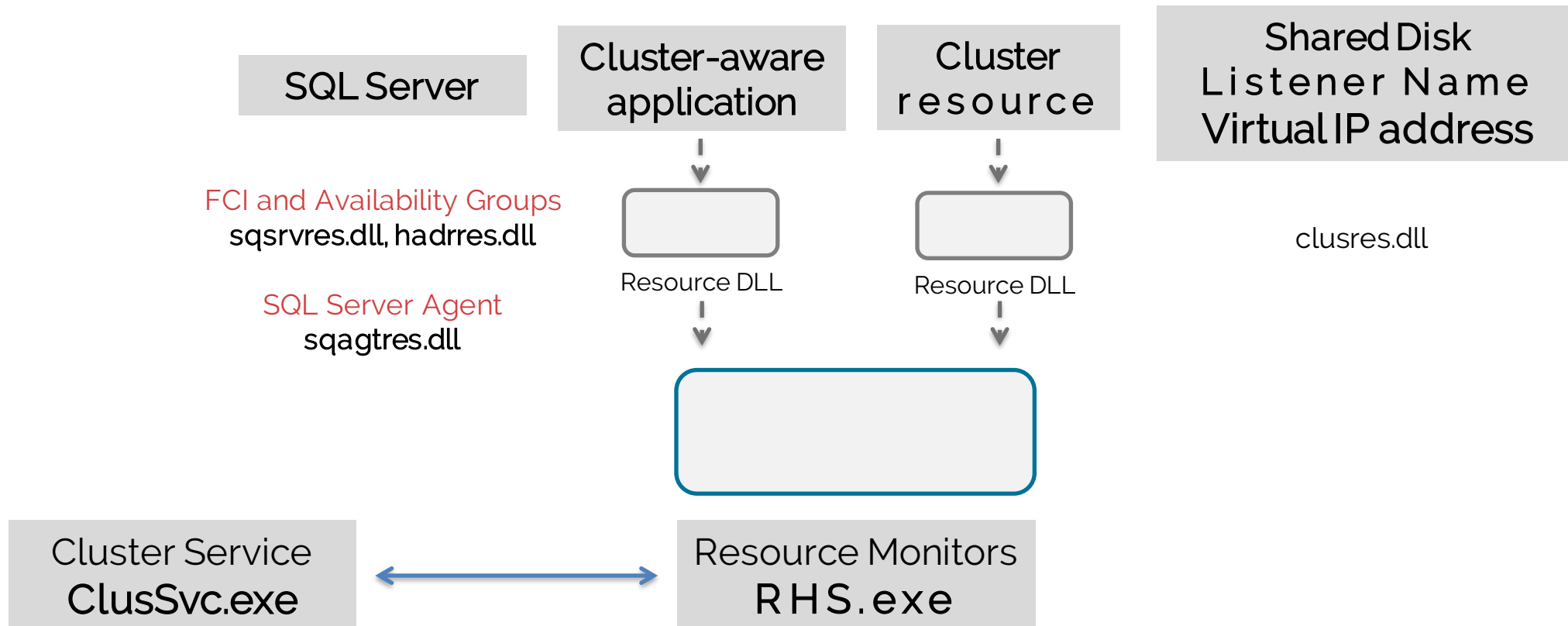


LooksAlive

checks SQL Server services every 5 seconds

IsAlive

runs **SELECT @@SERVERNAME** every 60 seconds
gets called immediately if **LooksAlive** fails
retries 5 times



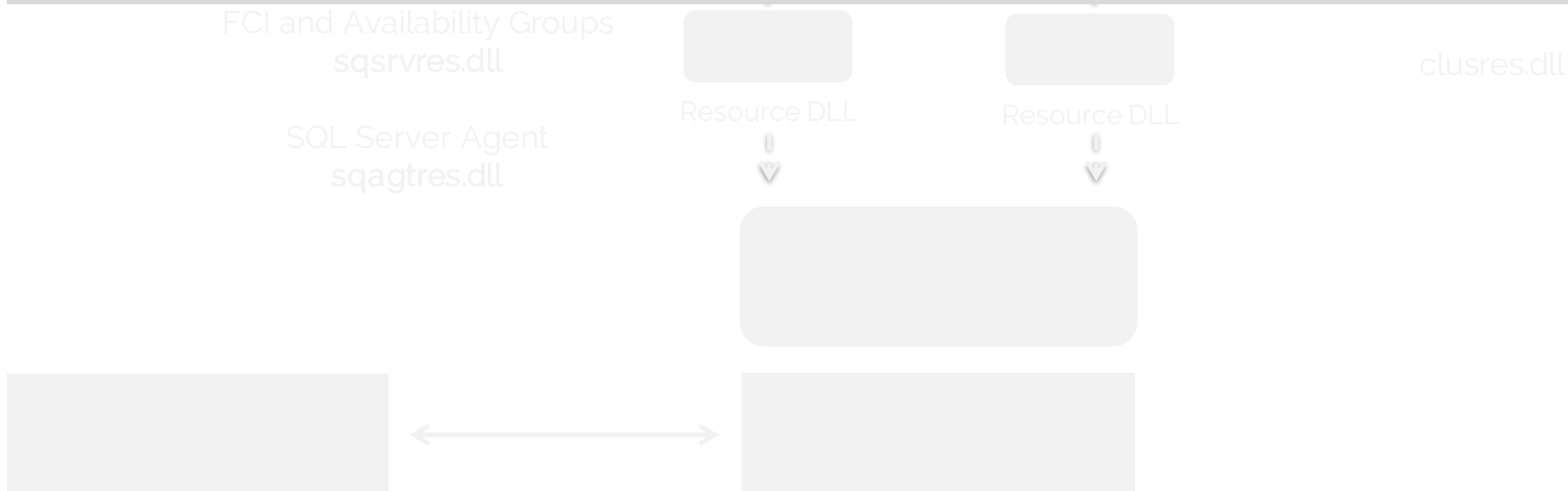
LooksAlive

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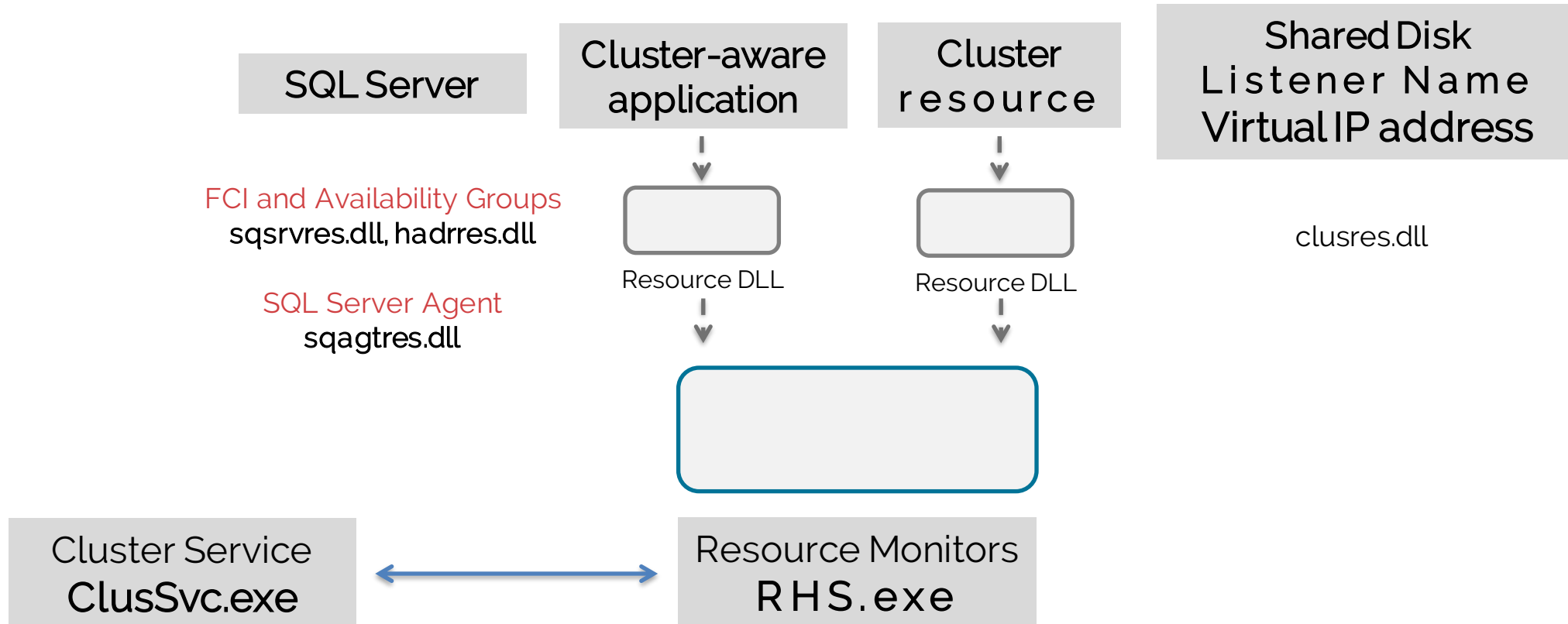
starting with **SQL Server 2012**



LooksAlive

IsAlive

sp_server_diagnostics



WSFC

INTERNALS

sp_server_diagnostics

captures diagnostic data and health information about SQL Server to detect potential failures

WSFC

INTERNALS

FailureConditionLevel

specifies what failure conditions trigger an automatic failover

WSFC

INTERNALS

HealthCheckTimeout

length of time (in ms) that the SQL Server resource DLL should wait for information returned by **sp_server_diagnostics**

WSFC

INTERNALS

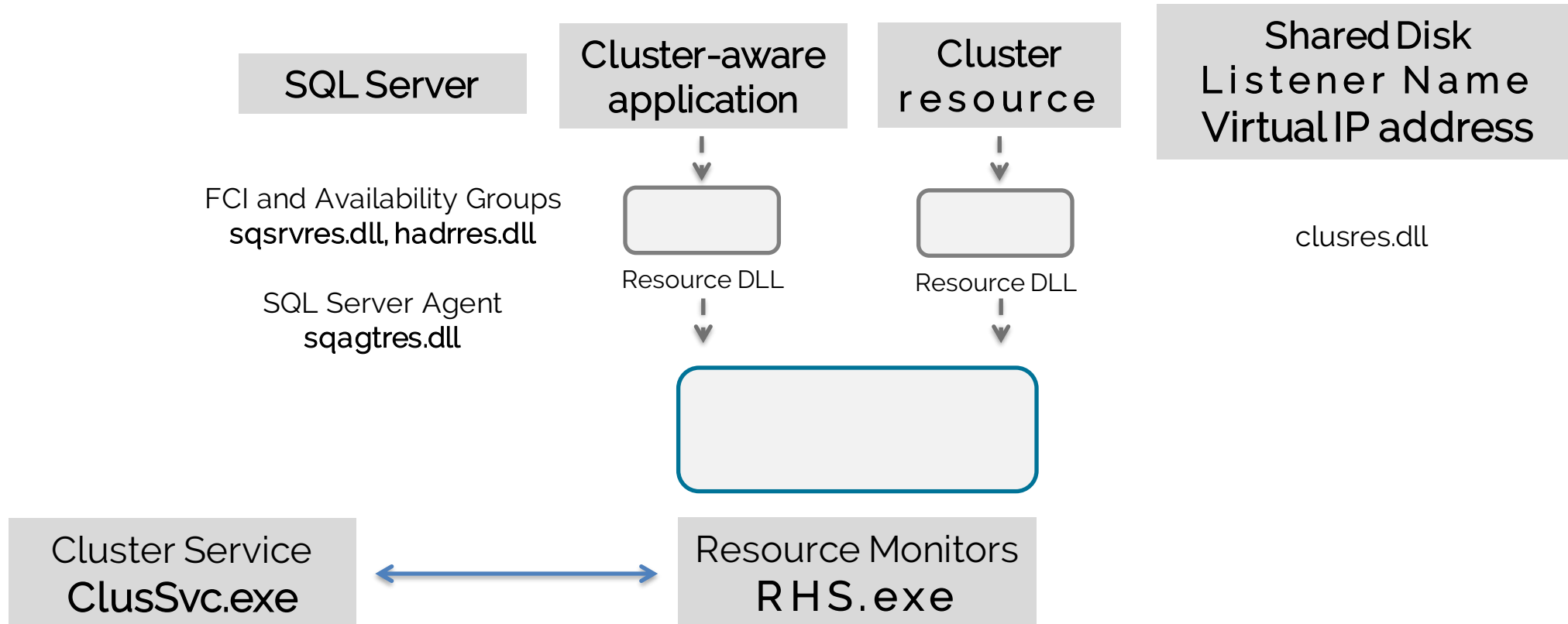
LeaseTimeout

controls the lease mechanism and
when the lease expires (in ms)

LooksAlive

IsAlive

sp_server_diagnostics



PUT IT ALL TOGETHER



Resource Monitors
RHS.exe



Cluster Service
ClusSvc.exe

WSFC

INTERNALS

IN ACTION

SQL Server

Cluster-aware
application

Cluster
resource

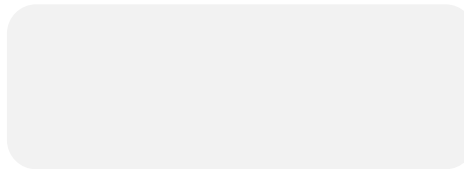
Shared Disk
Listener Name

CREATE AVAILABILITY GROUP [AG01] ...

Resource DLL



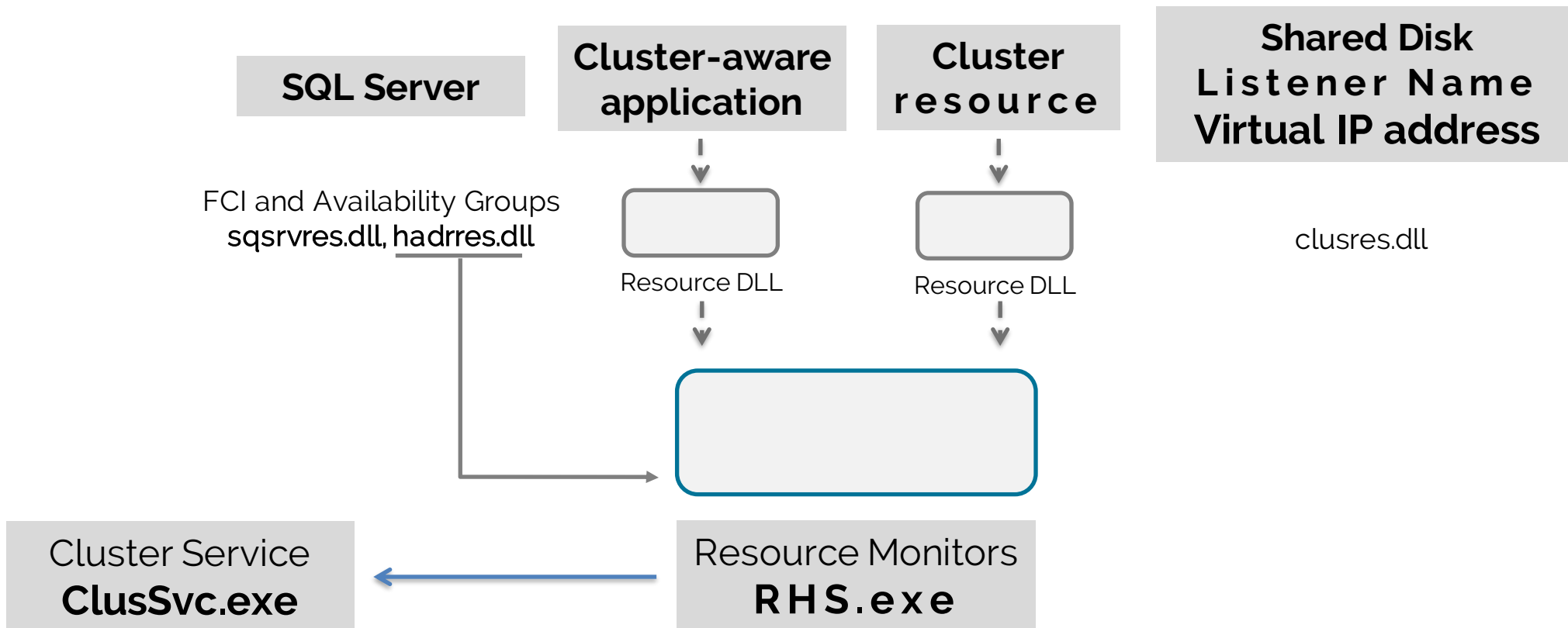
Resource DLL

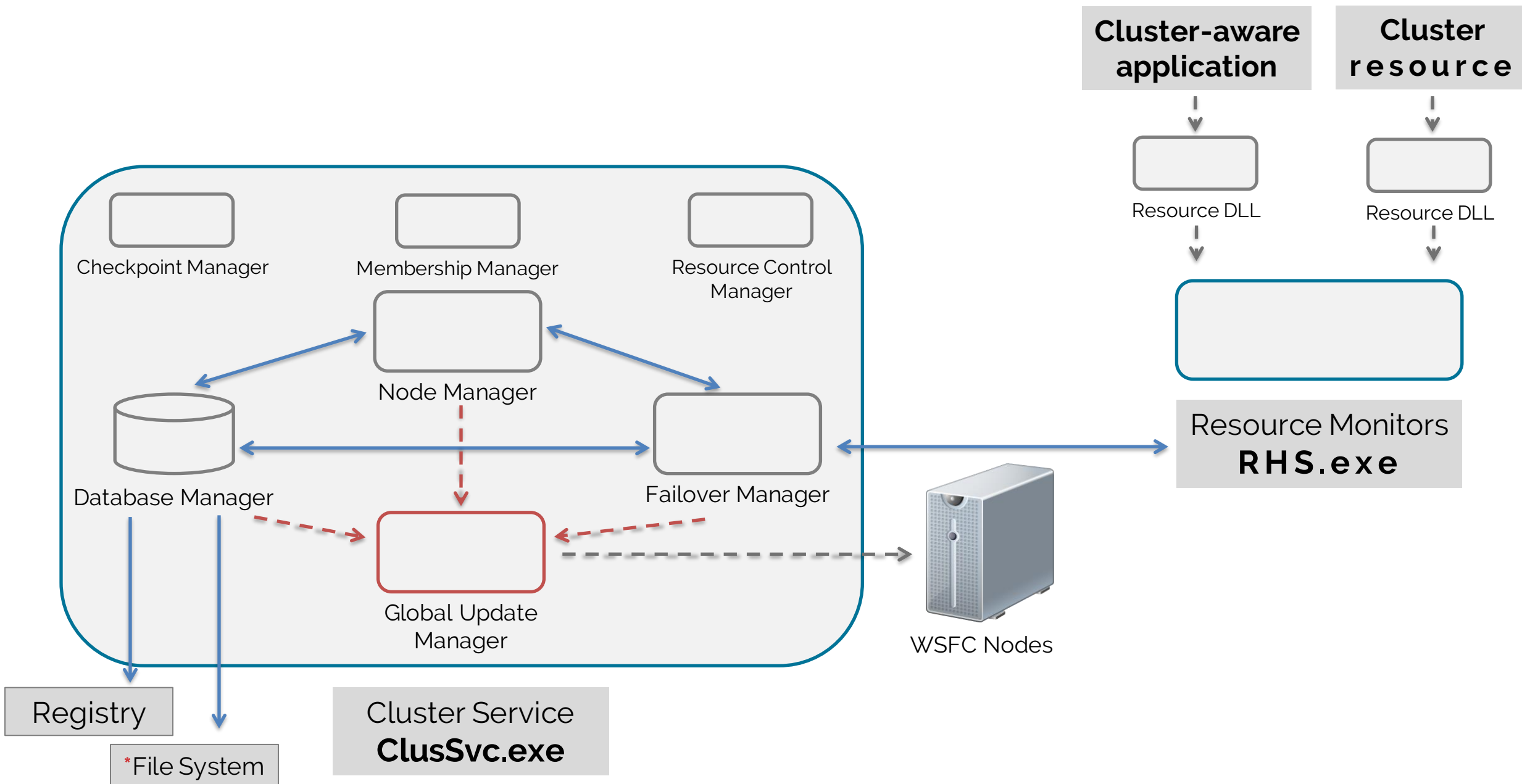


Cluster Service
ClusSvc.exe



Resource Monitors
RHS.exe





172.16.0.51 - Remote Desktop Connection

Failover Cluster Manager

File Action View Help

Failover Cluster Manager

TDPRDCLSAG025.TESTDOM

Roles

Nodes

Storage

Networks

Cluster Events

Roles (1)

Search

Queries

Name	Status	Type	Owner Node	Priority	Information
AG01	Pending	Other	TDPRD021	Medium	

Roles:

Windows Taskbar

System Tray

172.16.0.51 - Remote Desktop Connection

Failover Cluster Manager

File Action View Help

Failover Cluster Manager

- TDPRDCLSAG025.TESTDOM
 - Roles
 - Nodes
 - Storage
 - Networks
 - Cluster Events

Roles (1)

Search

Name	Status	Type	Owner Node	Priority	Information
AG01	Pending	Other	TDPRD021	Medium	

Roles:

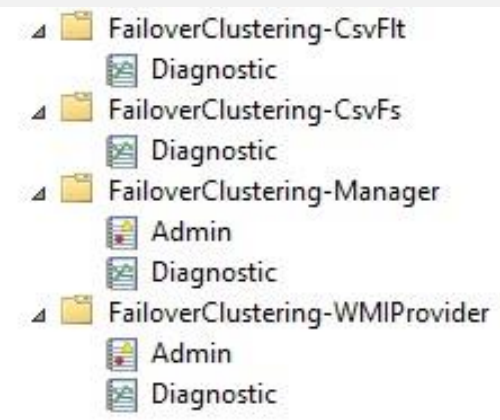
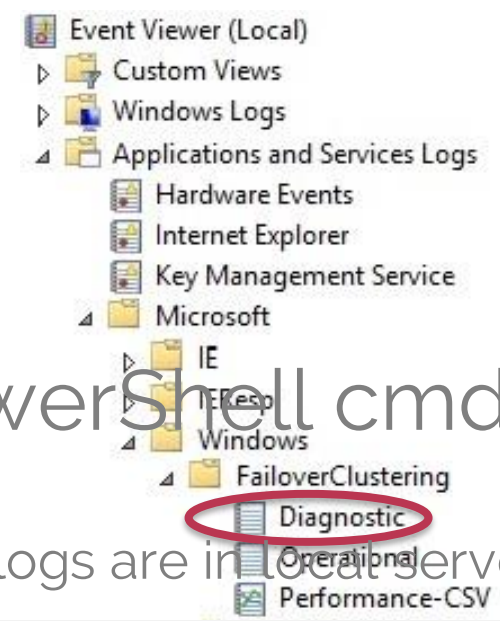
READING

CLUSTER ERROR LOGS

Get-ClusterLog PowerShell cmdlet

logs are in GMT, Windows event logs are in local server time

```
Get-ClusterLog -Destination . -TimeSpan 40 -UseLocalTime
```



INFO
WARN
ERR
DBG

Resource Name/API call

000012e4.00000674::2025/09/28-22:02:11.561 INFO [RES] SQL Server <SQL Server (TDPRDSQLFCI016)> [sqsrsvres] Request to bring SQL Server online

Thread
ID

Process
ID

Date/Time
(GMT)

Component

Description

“

A **BETTER WAY** TO ANALYZE THE LOGS



SQL LogScout

license MIT Stars 111 Forks 29 issues 1 open

<https://aka.ms/sqllogscout> gets you here

Note: For SQL Server on Linux, see [SQL LogScout Linux](#).

Table of Contents

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- [10. Security](#)
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<https://aka.ms/sqllogscout>

Troubleshoot SQL Always On Availability Groups with SQL LogScout

EDWIN SARMIENTO / APRIL 25, 2024 / AVAILABILITY GROUPS

Problem

We experienced several unplanned outages and failovers on our SQL Server Always On Availability Groups. We want to know the root cause to prevent them from happening in the future. How do we identify the root causes of unplanned Availability Group outages and failovers?

Solution

The goal of deploying an Always On Availability Group is to provide high availability to SQL Server databases. So, it becomes problematic when an unplanned outage or unexpected failover happens more often than necessary. You need to troubleshoot the issue and find the root cause to prevent it from happening again. The first step in troubleshooting an issue is to collect as many logs as possible for analysis.

AGDIAG

AGDiag diagnoses and reports failover and health events detected in the Cluster log of the primary re 

AGDIAG will detect and report on these issues:

- * Detect and analyze Cluster or SQL health issues that cause availability group to fail over or g
- * Detect and analyze Cluster or SQL health issues that cause SQL Failover Cluster Instance to fai
- * Detect and analyze why availability group failed to failover to failover partner during manual

HOW TO USE

<https://github.com/microsoft/agdiag>

- 1 Collect logs from the replica in the primary role when the health issue occurred using one of these 
 - SQL Log Scout, collect Scenario 0: Available on GitHub (https://github.com/microsoft/SQL_LogScout)
 - TSS, SQL Base Diagnostic collection: Available on GitHub (<https://github.com/walter-1/TSS>)
 - PSSDiag: Available on GitHub (<https://github.com/microsoft/DiagManager/releases/>)

- 2 Unzip the logs in a folder.

- 3 Launch AGDiag and select the log folder containing the unzipped logs.

Contributing

This project welcomes contributions and suggestions. Most contributions require you to agree to a Contributor License Agreement (CLA) declaring that you have the right to, and actually do, grant us the rights to use your contribution. For details, visit <https://cla.opensource.microsoft.com>.

Troubleshoot SQL Server Always On Availability Groups with AGDiag

EDWIN SARMIENTO / MAY 14, 2024 / AVAILABILITY GROUPS / 2 COMMENTS

Problem

Since we have experienced a number of unplanned outages and failovers on our SQL Server Always On Availability Groups, we want to know the root cause to prevent them from happening in the future. How do we identify the root causes of unplanned Availability Group outages and failovers?

Solution

In a previous tip, [Collect Logs for Analyzing Unplanned Availability Group Outages and Failovers using SQL LogScout](#), we collected the logs necessary to start analyzing unplanned Availability Groups outages and failovers. And while you can manually go through each log to decipher error messages and analyze potential root causes of the issues, there is an easier way to do so. The Microsoft technical support engineers created a tool called AGDiag that does exactly this. AGDiag is mentioned in [this Microsoft documentation](#) as the preferred tool for analyzing failover and failures of SQL

ADMIN TASKS

DO NOT DO THIS OUTSIDE OF SQL SERVER

PERFORMING ADMIN TASKS
USING FAILOVER CLUSTER MANAGER

*PLEASE INFORM YOUR SYSADMINS

ACTION ITEMS



WHAT'S
NEXT

20

master the FUNDAMENTALS of WSFC